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MILITARY SPECIFICATION

BREAD PUDDING, IMITATION MAPLE FLAVORED, WITH HAM, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers maple flavored bread pudding with ham in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-1497	-	Labeling of Metal Cans for Subsistence items.
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistance Items; Clothing, Personal Equipment and Equipage; General Specifications For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-900	-	Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars used in the Preparation of Thermostabilized Foods for the Armed Forces.

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR PARTS 1-199)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

(Copies may be obtained from the Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Washington, DC 20090-6456.)

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Meat and Poultry Inspection Regulations

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. Standards for Condition of Food Containers

(Copies may be obtained from the Chairman, Condition of Food Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0610, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies may be obtained from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Copies should be obtained from the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Copies should be obtained from the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

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NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Copies should be obtained from the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to the first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Ham, canned or cooked. The ham shall be high commercial style. The ham shall not be labeled "water added" and shall not contain dextrose. The canned or cooked ham shall be maintained in the temperature range of 28°F to 40°F for a period of time not to exceed 60 days from the date of pack until preparation of the finished product.

3.2.2 Eggs, whole, liquid or frozen. Whole eggs may be liquid or frozen and shall have been processed in accordance with Regulations Governing the Inspection of Eggs and Egg Products (7 CFR 59). The whole eggs shall be egg whites and egg yolks in their natural proportions as broken directly from the shell as evidenced by a USDA Egg Products Inspection Certificate. For liquid whole eggs, the certificate shall state the time of pasteurization. Liquid whole eggs shall be held at temperature of 40°F or lower and shall be held for

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not more than 72 hours from the time of pasteurization until the start of formulation of the product in which they are used. Frozen whole eggs shall be held at 10°F or lower and used within 120 days from the date of production. The whole eggs shall be free from off-odors, off-flavors, and foreign odors and flavors, such as metallic, musty, sour, or fruity and shall be free from foreign material.

3.2.3 Bread, white, enriched. Bread shall be commercial, soft crumb, enriched, white bread. The bread shall possess a flavor, aroma, and texture that is typical of soft crumb, white bread. The commercial white bread shall have a moisture content of 38.0 percent; however, if this moisture requirement is not met, the moisture can be adjusted to 38.0 percent as described in 3.3.3, footnote 1. The bread shall have a protein content of not less than 8.0 percent, a fat content of not less than 1.4 percent, and a carbohydrate content of not less than 50.0 percent.

3.2.4 Sugar, granulated. Sugar shall be white, refined, sucrose, granulated cane or beet, or a combination of both and shall comply with MIL-STD-900.

3.2.5 Oil, vegetable. Vegetable oil shall possess a clean, bland flavor, have a minimum stability of 15 hours (A.O.M.), have a free fatty acid value not to exceed 0.05 percent, have a moisture/volatile matter content not to exceed 0.06 percent, and pass a cold test of 5.5 hours (minimum).

3.2.6 Flavoring, maple, imitation. Imitation maple flavoring shall be a double strength product that possesses a color, flavor, and odor characteristic of maple flavoring (see 6.4).

3.2.7 Cream, dry. Dry cream shall comply with FDA Standards of Identity for dry cream except it shall contain 70 to 75 percent, by weight, of milk fat. The product shall contain no added nutritive sweeteners.

3.2.8 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.9 Water. Water used for formulation, ice making and washing shall conform to the National Primary Drinking Water Regulations.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Preparation of frozen whole eggs. Frozen whole eggs shall be tempered to 28°F to 40°F and held in the temperature range of 28°F to 40°F for not more than 24 hours prior to product preparation.

3.3.2 Preparation of ham. The ham shall be ground once through a plate having holes 0.5 inch in diameter. The ham shall be held for not more than 4 hours in the temperature range of 28°F to 40°F from the time of grinding until inclusion in the bread pudding.

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3.3.3 Preparation of bread pudding. The bread pudding shall be formulated and prepared as follows:

Ingredient	Percent by weight
Water	24.56
Eggs, whole	21.70
Ham, ground	23.00
Bread 1/	18.80
Sugar	9.00
Vegetable oil	1.56
Imitation maple flavor, double strength	0.40
Cream, dry	0.78
Lecithin	0.20

1/ The moisture content of the bread used in the formulation shall be determined before combining with the above ingredients. If the moisture content is determined to be greater than 38 percent, that amount in excess shall be deducted from the percent water in the formulation. If the moisture content is less than 38 percent, adjust the formulation with the required amount of water.

a. The dry cream shall be rehydrated in the water.

b. The balance of ingredients (except ham and bread) shall be combined with the rehydrated cream and mixed until the sugar is completely dissolved.

c. The bread shall be added and coarsely mixed to produce small but recognizable particles of bread.

d. The ground ham shall be added and mixed only for sufficient time to uniformly distribute the ham throughout the pudding.

e. Filling of the product into tray pack cans shall be a continuous operation. Filling shall be completed within 1 hour after product preparation. The time allowed from product preparation to can filling may be extended to 4 hours provided the internal product temperature does not exceed 40°F during this time. In addition, the mixture shall be periodically stirred during can filling to prevent settling.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1.) shall be filled with product so as to conform to the finished product requirements and to the following requirements.

a. Each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to assure compliance with the finished product requirements (see 3.61).

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b. The filled and sealed tray pack cans shall be in the retort process within 2 hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a minimum sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, glass, dirt, insect parts, hair, wood, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, moldy, rancid, sour, or stale.

c. There shall be no color foreign to the product.

d. The average net weight shall be not less than 104 ounces.

e. No individual can shall contain less than 102 ounces of product.

f. The product shall have a tan surface with a uniform creamy-tan interior, with discernable ham particles uniformly dispersed.

g. The product shall possess a slightly sweet maple flavor and moderate ham flavor.

h. The product shall have a smooth surface with a moist, dense, grainy custard interior and a spongy texture.

i. No individual tray pack shall have a fat content greater than 9.0 percent.

j. The average fat content shall be not greater than 7.0 percent.

k. The product shall show no evidence of excessive heating (materially darkened or scorched).

l. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

m. The product shall contain not more than one air pocket or void area measuring 1 inch or more in each of two separate dimensions.

n. The product shall contain not more than two air pockets or void areas measuring 1/2 inch or more in each of two separate dimensions.

o. No individual can shall contain a bone piece measuring 0.3 inch or more in any dimension.

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p. No individual can shall contain cartilage, coarse connective tissue, or section of tendons or ligaments measuring 0.3 inch or more in any dimension.

q. No individual can shall contain skin, blood clot, bruise, or discolored meat measuring 0.3 inch or more in any dimension.

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods," and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or other applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists to determine conformance to the requirements. Any nonconformance to any identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review or records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 104 ounces or if the net weight of any individual can is less than 102 ounces, the lot shall be rejected.

4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated for 35 minutes in boiling water, opened, and inspected for the defects listed in table II.

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TABLE I. Double sampling plan for product inspection. 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3200	8	---	0	2
	8	16	1	2
3201 to 35000	13	---	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/ 3/

Category	Defect
Major	
101	Product surface not tan, interior not a uniform creamy-tan, and discernable ham particles not uniformly dispersed.
102	Product does not possess a slightly sweet maple flavor and a moderate ham flavor.
103	Product does not possess a smooth surface with a moist, dense, grainy custard interior, and spongy texture.
104	Evidence of excessive heating (materially darkened or scorched).
105	Presence of two or more air pockets or void areas measuring 1 inch or more in each of two separate dimensions 4/.
106	Presence of three or more air pockets or void areas measuring 1/2 inch or more in each of two separate dimensions 4/.
107	Presence of a bone piece measuring 0.3 inch or more in any dimension.

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TABLE II. Product defects 1/ 2/ 3/ (cont'd)

Category	Defect
Major	
108	Presence of cartilage, coarse connective tissue, or section of tendons or ligaments measuring 0.3 inch or more in any dimension.
109	Presence of skin, blood clot, bruise, or discolored meat measuring 0.3 inch or more in any dimension.

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ The product shall be divided into ten approximately equal portions and examined for appearance and texture characteristics.

4/ From each sample can of product, remove one 3 inch wide center slice (sliced lengthwise of the can). Place center slice on edge and cut in half lengthwise. Inspect right inside surface for air pockets and void areas.

4.5.3.3 Fat content testing. Nine tray packs shall be selected at random from the lot and distributed as follows:

- Three for laboratory analysis
- Three for submission to the contractor
- Three for retention by an AMS agent as a reserve sample.

The three tray packs for laboratory analysis shall individually be tested for fat content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists: Chapter: Cereal Foods, except that preparation of the samples shall be as follows: The three unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. Test results shall be reported to the nearest 0.1 percent. Any test failure is a major defect. The lot shall be rejected if any of the following major defects occur:

a. The average fat content of the three tray packs is greater than 7.0 percent.
b. The fat content of any individual tray pack is greater than 9.0 percent.

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Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat content requirement. When the reserve samples are analyzed, the analysis for fat shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers except that inspection for labeling shall be as specified in 4.5.4.1.

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4), except that for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can, i.e., label raised or peeled back from edges or corners.

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D-3330.

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and CFR 21, Part 113, Subpart D, or CFR 9, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns is a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 75 ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. When examining a ribbed lid, only lay the straight edge between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance is a major defect and shall be cause for rejection of the lot.

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4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 104 ounces of the food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify the compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.1.1 When specified (see 6.1), the base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 98 pounds per base box. The body design shall be reinforced on the bottom with six longitudinal beads or ribs 0.050 ± 0.010 inch deep. The beads shall be 9 inches by 3/4 inch in length and width and shall be equidistant in the center panel of the bottom. The end design shall be reinforced with eight longitudinal beads 0.018 ± 0.004 inch deep. The beads shall be 11 inches by 3/4 inch in length and width and shall be equidistant in the center panel of the end. The center four beads on the end design shall be interrupted to provide a 5-inch by 5-inch flat center panel for labeling. The apex of all these beads shall be outward. The interior and exterior can coating requirements shall be as specified in 5.1.1.

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5.1.2 Level C. The product shall be preserved as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with a HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1) except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1 except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box with the first two cans placed with the lids together and the next two cans also with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.4 Labeling. Each tray pack shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (e.g. EST 38) or a three letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as four digit Julian code. The first digit shall indicate the year of production, and the next three digits shall indicate the day of the year (Example, March 19, 1987 would be coded as 7078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

Alternatively, a preprinted self-adhering 0.002 inch thick clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

"Tray cans may be heated prior to serving or served at ambient temperature without heating. When heated, the following heating directions shall apply:

TO HEAT: Submerge unopened can in boiling water. Simmer gently 40 - 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 35 - 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions cut 3 rows by 6 rows"

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5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or shipping order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, when tray pack cans are fabricated from 90 pounds per base box, the following precautionary marking in capital letters, larger than other markings, shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4 and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of preservation and packing required (see 5.1 and 5.2).
e. When tray pack cans are to be fabricated from 98 pounds per base box (see 5.1.1.1).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article inspection. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Flavoring, maple, imitation. It has been found that Durkee's Standard Imitation Maple Flavoring (Durkee Industrial Foods, 925 Euclid Avenue, Cleveland, OH 44155) and Cunningham Imitation Maple Flavor No. S-L-102 (Dirigo Corporation, P.O. Box E4, 141 W. 2nd Street, South Boston, MA 02127) satisfactorily colors and flavors the product such as to meet the requirements in 3.6 f and g.

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6.5 Subject term (key word) listing.

Bread pudding
Canned food
Thermostabilized
Tray pack

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8940-0598

Reveiw activities:

Army - MD, TS
Navy - MC
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44249A	Bread Pudding, Imitation Maple Flavored, with Ham, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP-1, Feb 86

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